

Probiotics for the prevention of pediatric antibiotic-associated diarrhea.

BACKGROUND: Antibiotics alter the microbial balance within the gastrointestinal tract. Probiotics may prevent antibiotic-associated diarrhea (AAD) via restoration of the gut microflora. Antibiotics are prescribed frequently in children and AAD is common in this population. **OBJECTIVES:** To assess the efficacy and adverse effects of probiotics (any specified strain or dose) for the prevention of antibiotic-associated diarrhea in children. To assess adverse events associated with the use of probiotics when co-administered with antibiotics in children. **SEARCH STRATEGY:** MEDLINE, EMBASE, CENTRAL, CINAHL , AMED, and the Web of Science (inception to August 2006) were searched along with specialized registers including the Cochrane IBD/FBD Review Group, CISCOR, Chalmers PedCAM Research Register and trial registries from inception to 2005. Letters were sent to authors of included trials, nutra/pharmaceutical companies, and experts in the field requesting additional information on ongoing or unpublished trials. Conference proceedings, dissertation abstracts, and reference lists from included and relevant articles were hand searched. **SELECTION CRITERIA:** Randomized, parallel, controlled (placebo, active, or no treatment) trials comparing co-administered probiotics with antibiotics for the prevention of diarrhea secondary to antibiotic use in children (0 to 18 years). **DATA COLLECTION AND ANALYSIS:** Methodological quality assessment and data extraction were conducted independently by two authors (BCJ, AS). Dichotomous data (incidence of diarrhea, adverse events) were combined using pooled relative risks, and continuous data (mean duration of diarrhea, mean daily stool frequency) as weighted mean differences, along with their corresponding 95% confidence intervals. Adverse events were summarized using risk difference. For overall pooled results on the incidence of diarrhea, a priori sensitivity analyses included per protocol versus intention to treat, random versus fixed effects, and methodological quality criterion. Subgroup analysis were conducted on probiotic strain, dose, definition of antibiotic-associated diarrhea, and antibiotic agent. **MAIN RESULTS:** Ten studies met the inclusion criteria. Trials included treatment with either *Lactobacilli* spp., *Bifidobacterium* spp., *Streptococcus* spp., or *Saccharomyces boulardii* alone or in

combination. Six studies used a single strain probiotic agent and four combined two probiotic strains. The per protocol analysis for 9/10 trials reporting on the incidence of diarrhea show statistically significant results favouring probiotics over active/non active controls (RR 0.49; 95% CI 0.32 to 0.74). However, intention to treat analysis showed non-significant results overall (RR 0.90; 95% CI 0.50 to 1.63). Five of ten trials monitored for adverse events (n = 647); none reported a serious adverse event.

AUTHORS' CONCLUSIONS: Probiotics show promise for the prevention of pediatric AAD. While per protocol analysis yields treatment effect estimates that are both statistically and clinically significant, as does analysis of high quality studies, the estimate from the intention to treat analysis was not statistically significant. Future studies should involve probiotic strains and doses with the most promising evidence (e.g., *Lactobacillus* GG, *Lactobacillus sporogenes*, *Saccharomyces boulardii* at 5 to 40 billion colony forming units/day). Research done to date does not permit determination of the effect of age (e.g., infant versus older children) or antibiotic duration (e.g., 5 days versus 10 days). Future trials would benefit from a validated primary outcome measure for antibiotic-associated diarrhea that is sensitive to change and reflects what treatment effect clinicians, parents, and children consider important. The current data are promising, but it is premature to routinely recommend probiotics for the prevention of pediatric AAD.